Further Calculus Topic Test 1 Solution

Total Marks: 30

Time: 50 minutes

1 Which of the following is an expression for $\int sin^2 4x\, dx$? 1

A. $\frac{x}{2} - \frac{1}{16} sin8x + C$

B. $\frac{x}{2} + \frac{1}{16} sin8x + C$

C. $x - \frac{1}{8} sin8x + C$

D. $x + \frac{1}{8} sin8x + C$

$\int \sin^2 4x\, dx$

$= \frac{1}{2}\int (1 - \cos 8x)\, dx$

$= \frac{1}{2}\left(x - \frac{1}{8}\sin 8x\right) + C$ $\boxed{A}$

$= \frac{x}{2} - \frac{1}{16}\sin 8x + C$

2 Find $\int tanx\ \ dx$. 1

$\int \tan x\, dx = -\int \frac{-\sin x}{\cos x}\, dx$

$= -\ln|\cos x| + C = \ln|\sec x| + C$

3 Evaluate $\int_{\frac{\pi}{2}}^{\frac{3\pi}{4}} cos^2 2\theta\, sin2\theta\ \ d\theta$. Use the substitution $u = cos2\theta$. 3

$\int_{\frac{\pi}{2}}^{\frac{3\pi}{4}} \cos^2 2\theta \sin 2\theta\, d\theta$

$= \int_{-1}^{0} -\frac{1}{2}u^2\, du$

$= -\frac{1}{2}\int_{-1}^{0} u^2\, du = -\frac{1}{2}\left[\frac{u^3}{3}\right]_{-1}^{0}$

$= -\frac{1}{2}\left(0 - -\frac{1}{3}\right) = -\frac{1}{6}$

$u = \cos 2\theta$

$\frac{du}{d\theta} = -2\sin 2\theta$

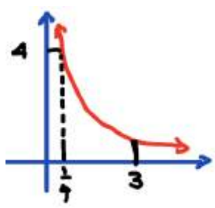
$du = -2\sin 2\theta\, d\theta$

$\sin 2\theta\, d\theta = -\frac{1}{2}\, du$

when $\theta = \frac{\pi}{2}$ $u = \cos\pi = -1$

when $\theta = \frac{3\pi}{4}$ $u = \cos\frac{3\pi}{2} = 0$

4 A solid is generated when the region bounded by $y = \frac{1}{x}$, $y = 4$, the line $x = 3$ and the $y -$ axis is rotated about the $x -$ axis. Determine the volume of the solid. 3



The solid is made up of a cylinder of radius 4 and height $\frac{1}{4}$ and a solid formed when the area under the curve $y = \frac{1}{x}$ $\frac{1}{4} \leq x \leq 3$ is rotated.

$V = \left(\pi 4^2 \times \frac{1}{4}\right) + \pi\int_{\frac{1}{4}}^{3} \frac{1}{x^2}\, dx = 4\pi + \pi\int_{\frac{1}{4}}^{3} x^{-2}\, dx$

$= 4\pi + \pi\left[\frac{x^{-1}}{-1}\right]_{\frac{1}{4}}^{3} = 4\pi + \pi\left[-\frac{1}{x}\right]_{\frac{1}{4}}^{3}$

$= 4\pi + \pi\left(-\frac{1}{3} - -4\right) = 4\pi + \frac{11\pi}{3} = \frac{23\pi}{4}$ units^3

5 (i) Find $\int \frac{x}{\sqrt{1-x^2}}dx$ using the substitution $u = 1 - x^2$. 2

(ii) Differentiate $xcos^{-1}x$ with respect to x. 2

(iii) Hence, find $\int cos^{-1}x\,dx$. 2

i) Let $u = 1-x^2$

$\frac{du}{dx} = -2x$

$-\frac{1}{2}du = x\,dx$

$\int \frac{x}{\sqrt{1-x^2}}dx = -\frac{1}{2}\int \frac{1}{\sqrt{u}}du$

$= -\frac{1}{2}\int u^{-\frac{1}{2}}dy$

$= -\frac{1}{2}\left(2u^{\frac{1}{2}}\right) + C = -\sqrt{1-x^2} + C$

ii) $u = x \qquad v = \cos^{-1}x$

$u' = 1 \qquad v' = -\frac{1}{\sqrt{1-x^2}}$

$\frac{d}{dx}\left(x\cos^{-1}x\right) = \cos^{-1}x - \frac{x}{\sqrt{1-x^2}}$

iii) $\int \cos^{-1}x - \frac{x}{\sqrt{1-x^2}}dx = x\cos^{-1}x$

$\int \cos^{-1}x\,dx = x\cos^{-1}x + \int \frac{x}{\sqrt{1-x^2}}dx$

$\int \cos^{-1}x\,dx = x\cos^{-1}x - \sqrt{1-x^2} + C$

6 Use the substitution $u = 2e^x$ to show that $\int_{ln\left(\frac{1}{2}\right)}^{ln\left(\frac{\sqrt{3}}{2}\right)} \frac{e^x}{1+4e^{2x}}dx = \frac{\pi}{24}$. 3

$I = \int_{\ln\frac{1}{2}}^{\ln\frac{\sqrt{3}}{2}} \frac{e^x}{1+4e^{2x}}dx$

Let $u = 2e^x$

$du = 2e^x\,dx$

$x = \ln\frac{\sqrt{3}}{2}, \quad u = \sqrt{3}$

$x = \ln\frac{1}{2}, \quad u = 1$

$I = \frac{1}{2}\int_{\ln\frac{1}{2}}^{\ln\frac{\sqrt{3}}{2}} \frac{2e^x}{1+(2e^x)^2}dx$

$= \frac{1}{2}\int_1^{\sqrt{3}} \frac{du}{1+u^2}$

$= \frac{1}{2}\left[\tan^{-1}u\right]_1^{\sqrt{3}}$

$= \frac{1}{2}\left[\tan^{-1}\sqrt{3} - \tan^{-1}1\right]$

$= \frac{1}{2}\left(\frac{\pi}{3} - \frac{\pi}{4}\right) = \frac{\pi}{24}$

7 Given $y = xtan^{-1}(2x)$, find y'. 2

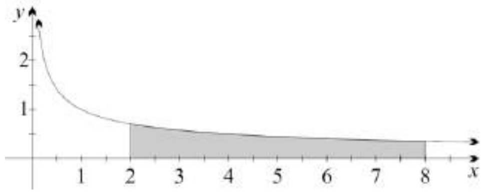
$u = x \qquad v = \tan^{-1}(2x)$

$u' = 1 \qquad v' = \frac{2}{1+4x^2}$

$y' = vu' + uv'$

$= \tan^{-1}(2x) + \frac{2x}{1+4x^2}$

8 The diagram above shows the region enclosed by the graph of $y = \frac{1}{\sqrt{x}}$, the vertical lines $x = 2$ and $x = 8$, and the $x -$ axis. This region is rotated about the x-axis. Find the exact volume of the solid created. Express your answer in simplified form. 3



$y = \frac{1}{\sqrt{x}}$,
$y^2 = \frac{1}{x}$

$\int_2^8 \pi y^2 dx = \pi \int_2^8 \frac{1}{x} dx$
$= \pi \left[\ln|x| \right]_2^8$
$= \pi \left[\ln(8) - \ln(2) \right] = \pi \ln 4$

9 Find $\int \frac{1}{9+16x^2} dx$. 2

$\int \frac{1}{9+16x^2} dx$

from ref. sheet: $\int \frac{f'(x)}{a^2+[f(x)]^2} dx = \frac{1}{a} \tan^{-1} \frac{f(x)}{a} + c$

$\int \frac{1}{3^2+[4x]^2} dx$

$= \frac{1}{4} \int \frac{4}{3^2+[4x]^2} dx = \frac{1}{4} \times \frac{1}{3} \tan^{-1}\left(\frac{4x}{3}\right) + c$

$= \frac{1}{12} \tan^{-1}\left(\frac{4x}{3}\right) + c$

10 Find $\frac{d}{dx} cos^{-1}\left(\frac{1}{x} - 1\right)$, expressing your answer in the form $\frac{1}{p\sqrt{q-1}}$ where p and q are expressions in terms of x. You may assume $x > \frac{1}{2}$. 3

$\frac{d}{dx}\left(\cos^{-1}\left(\frac{1}{x} - 1\right)\right) = \frac{1}{p\sqrt{q-1}}$

if $y = \cos^{-1} f(x)$, $\frac{dy}{dx} = \frac{-f'(x)}{\sqrt{1-[f(x)]^2}}$

$y = \cos^{-1}\left(\frac{1}{x} - 1\right)$
$= \cos^{-1}(x^{-1} - 1)$

$\frac{dy}{dx} = \frac{-(-x^{-2})}{\sqrt{1-\left[\frac{1}{x}-1\right]^2}} = \frac{\frac{1}{x^2}}{\sqrt{1-\left[\frac{1-x}{x}\right]^2}} = \frac{\frac{1}{x^2}}{\sqrt{1-\left[\frac{1-x}{x}\right]^2}}$

$= \frac{\frac{1}{x^2}}{\sqrt{\frac{x^2}{x^2} - \frac{(1-x)^2}{x^2}}}$ (or any valid simplification). ✓

$= \frac{\frac{1}{x^2}}{\sqrt{\frac{1}{x^2}\left(x^2-(1-x)^2\right)}}$

$= \frac{\frac{1}{x^2}}{\frac{1}{x}\sqrt{x^2-(1-2x+x^2)}} = \frac{1}{x\sqrt{2x-1}}$

11 Use double angle formulae to find $\int \sin^2 x \cos^2 x \, dx$. 2

$\int \sin^2 x \cos^2 x dx = \int \left(\frac{1}{2} - \frac{1}{2}\cos(2x)\right)\left(\frac{1}{2} + \frac{1}{2}\cos(2x)\right) dx$

$= \frac{1}{4} \int 1 - \cos^2(2x) dx$

$= \frac{1}{4} \int \frac{1}{2} - \frac{1}{2}\cos(4x) dx$ ✓

$= \frac{1}{4}\left(\frac{1}{2}x - \frac{1}{8}\sin(4x)\right) + C$

$= \frac{1}{8}x - \frac{1}{32}\sin(4x) + C$ ✓

12 Which expression is equal to $\int cos^2 \frac{2x}{5}$? 1

A. $\frac{x}{2} - \frac{2}{5}\sin\frac{4x}{5} + c$

B. $\frac{x}{2} + \frac{2}{5}\sin\frac{4x}{5} + c$

C. $\frac{x}{2} - \frac{5}{8}\sin\frac{4x}{5} + c$

D. $\frac{x}{2} + \frac{5}{8}\sin\frac{4x}{5} + c$

$\cos^2 x = \frac{1}{2} + \frac{1}{2}\cos 2x$.

$\cos^2\left(\frac{2x}{5}\right) = \frac{1}{2} + \frac{1}{2}\cos\left(2 \times \frac{2x}{5}\right)$

$\cos^2\left(\frac{2x}{5}\right) = \frac{1}{2} + \frac{1}{2}\cos\left(\frac{4x}{5}\right)$

$\int \left(\frac{1}{2} + \frac{1}{2}\cos\left(\frac{4x}{5}\right)\right) dx$

$= \frac{1}{2}x + \frac{1}{2} \times \frac{5}{4} \sin \frac{4x}{5}$

$= \frac{1}{2}x + \frac{5}{8}\sin\frac{4x}{5} + C$

Further Calculus Topic Test 2
Solution

Total Marks: 30

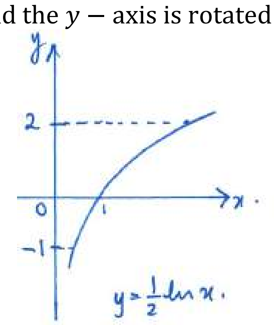
Time: 50 minutes

1 Find $\int \frac{4}{25+16x^2}dx$. $\quad \frac{1}{5}tan^{-1}\frac{4x}{5}+C$ 1

2 Evaluate $\int \frac{2x}{\sqrt{25-x^2}}dx$. 2

$$\int \frac{2x}{\sqrt{25-x^2}}dx = \int 2x\cdot(25-x^2)^{-\frac{1}{2}}dx$$
$$= -\frac{1}{\frac{1}{2}}(25-x^2)^{\frac{1}{2}}+C \quad \text{(reverse chain rule)}$$
$$= -2\sqrt{25-x^2}+C$$

3 Find the volume, in terms of π, of the solid formed when the area bounded by the curve $y=\frac{1}{2}log_e x$, the lines $y=-1$, $y=2$ and the $y-$axis is rotated about the $y-$axis. 3



$y = \frac{1}{2}\ln x$

$2y = \ln x$

$e^{2y} = x$

$$V = \pi\int_a^b x^2\,dy$$
$$= \pi\int_{-1}^{2}(e^{2y})^2\,dy$$
$$= \pi\int_{-1}^{2} e^{4y}\,dy$$
$$= \left[\frac{e^{4y}}{4}\right]_{-1}^{2}\times\pi$$
$$V = \left(\frac{e^8}{4}-\frac{e^{-4}}{4}\right)\pi \text{ units}^3.$$

4 Evaluate the integral $\int_0^{\frac{1}{2}}\frac{sin^{-1}x}{\sqrt{1-x^2}}dx$. $\quad \frac{\pi^2}{72}$ 2

5 Evaluate the integral $\int_0^{\sqrt{5}}\frac{2x^3}{\sqrt{x^2+4}}dx$ using the substitution $u=x^2+4$. 4

$$I = \int_0^{\sqrt{5}}\frac{2x^3}{\sqrt{x^2+4}}dx \quad [u = x^2+4] \qquad \frac{du}{dx} = 2x \;\therefore du = 2x\,dx. \text{ Also, } x^2 = u-4$$

when $\begin{cases} x=0, & u=4 \\ x=\sqrt{5}, & u=9. \end{cases}$

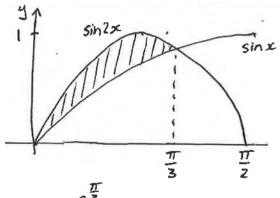
$$I = \int_4^9 \frac{u-4}{\sqrt{u}}du = \int_4^9 (u^{\frac{1}{2}} - 4u^{-\frac{1}{2}})du$$
$$= \left[\frac{2}{3}u^{\frac{3}{2}} - 8u^{\frac{1}{2}}\right]_4^9 = \left[\frac{2}{3}(27) - 8(3)\right] - \left[\frac{2}{3}(8) - 8(2)\right]$$
$$= 18 - 24 - \frac{16}{3} + 16 = \frac{14}{3}$$

6 Find $\int \frac{2tan5\theta}{sec^2 5\theta}d\theta$. 2

$$I = \int 2\frac{\sin 5\theta}{\cos 5\theta}\cdot\cos^2 5\theta\,d\theta = \int 2\sin 5\theta\cos 5\theta\,d\theta$$
$$= \int \sin 10\theta\,d\theta = -\frac{1}{10}\cos 10\theta + C$$

7 The region enclosed by $y = sinx$ and $y = sin2x$ in the domain $0 \le x \le \frac{\pi}{2}$ is rotated about the $x-$axis. 2

Calculate the volume of the resulting solid of revolution.



$\sin 2x = \sin x$

$\therefore\ 2\sin x\cos x = \sin x$

$\sin x(2\cos x - 1) = 0$

$\therefore\ x = 0 \text{ or } \frac{\pi}{3}$

$$V = \pi\int_0^{\frac{\pi}{3}} (\sin^2 2x - \sin^2 x)\,dx$$

$$= \pi\int_0^{\frac{\pi}{3}} \tfrac{1}{2}(1-\cos 4x) - \tfrac{1}{2}(1-\cos 2x)\ dx$$

$$= \frac{\pi}{2}\int_0^{\frac{\pi}{3}} (\cos 2x - \cos 4x)\,dx$$

$$V = \frac{\pi}{2}\left[\frac{\sin 2x}{2} - \frac{\sin 4x}{4}\right]_0^{\frac{\pi}{3}}$$

$$= \frac{\pi}{2}\left[\frac{\sqrt{3}}{4} - \left(-\frac{\sqrt{3}}{8}\right)\right] = \frac{\pi}{2}\left[\frac{3\sqrt{3}}{8}\right] = \frac{3\sqrt{3}\,\pi}{16} \text{ units}^3.$$

8 The function $y = f(x)$ is odd, and $f(x) \ge 0$ for $x \ge 0$. $\int_0^a f(x)\,dx = e^a + \frac{1}{e^a} - 2$ if $a > 0$. Find the area bounded by $y = f(x+4)$ and the straight lines $x = 5, x = -5$, and the $x-$axis. (Leave answer in exact form). 2

$y = f(x)$ is odd, $f(x) \ge 0$ for $x \ge 0$.

$$\text{Area} = \int_{-5}^{5} f(x+4)\,dx$$

Let $u = x+4$, $du = dx$, $x=5 \to u=9$, $x=-5 \to u=-1$

$$= \int_{-1}^{9} f(u)\,du$$

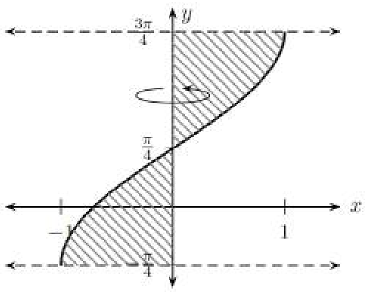
$$= \int_{-1}^{0} f(u)\,du + \int_{0}^{9} f(u)\,du$$

$$= \underbrace{\int_{0}^{1} f(u)\,du}_{\text{as } f(x) \text{ is odd}} + \int_{0}^{9} f(u)\,du$$

$$= \left(e^1 + \frac{1}{e^1} - 2\right) + \left(e^9 + \frac{1}{e^9} - 2\right)$$

$$\therefore\ \text{Area} = e + \frac{1}{e} + e^9 + \frac{1}{e^9} - 4 \quad \text{units}^2 \ \#$$

9 The region bounded by the curve $f(x) = sin^{-1}(x) + \frac{\pi}{4}$ and the $y-$axis between the lines $y = -\frac{\pi}{4}$ and $y = \frac{3\pi}{4}$, is rotated one complete revolution about the $y-$axis.



(i) Show that $sin^2\left(x - \frac{\pi}{4}\right) = \frac{1}{2}(1 - sin2x)$. 2

(ii) Hence, or otherwise, find the volume of the solid of revolution. Give your answer as an exact value. 2

i. $\text{LHS} = \left(\sin x\cos\frac{\pi}{4} - \cos x\sin\frac{\pi}{4}\right)^2$

$= \frac{1}{2}(\sin x - \cos x)^2$

$= \frac{1}{2}(\sin^2 x + \cos^2 x - 2\sin x\cos x)$

$= \frac{1}{2}(1 - \sin 2x) = \text{RHS}$

ii. $y = \sin^{-1}(x) - \frac{\pi}{4}$

$y + \frac{\pi}{4} = \sin^{-1}(x)$

$x = \sin\left(y + \frac{\pi}{4}\right)$

Let V be the volume of the solid of revolution about the y-axis:

$$V = \pi\int_{-\frac{\pi}{4}}^{\frac{3\pi}{4}} \left[\sin\left(y - \frac{\pi}{4}\right)\right]^2 dy$$

From part (i):

$$= \pi^2\int_{-\frac{\pi}{4}}^{\frac{3\pi}{4}} \frac{1}{2}(1 - \sin 2x)\,dy$$

$$= \frac{\pi^2}{2}\left[x + \frac{1}{2}\cos 2x\right]_{-\frac{\pi}{4}}^{\frac{3\pi}{4}}$$

$$= \frac{\pi^2}{2}\left[\left(\frac{3\pi}{4} + 0\right) - \left(-\frac{\pi}{4} + 0\right)\right] = \frac{\pi^2}{2}$$

10 Use the substitution $u = 3 + e^x$ to find the exact value of $\int_0^{\frac{\pi}{6}} \frac{e^x}{\sqrt{3+e^x}}\,dx$. 2

$$u = 3 + e^x \to du = e^x dx \text{ and } u_1 = 3 + e^{\ln 6} = 9 \text{ and } u_2 = 3 + e^0 = 4$$

$$\int_0^{\ln 6} \frac{e^x}{\sqrt{3+e^x}}\,dx = \int_4^9 \frac{du}{\sqrt{u}} = 2\left[\sqrt{u}\right]_4^9 = 2\left(\sqrt{9} - \sqrt{4}\right) = 2$$

11 Evaluate $\int_0^{\frac{\pi}{2}} cos^2 2x\, dx$. 2

$$\int_0^{\frac{\pi}{2}} \cos^2 2x\, dx = \frac{1}{2}\int_0^{\frac{\pi}{2}} (1+\cos 4x)\, dx$$

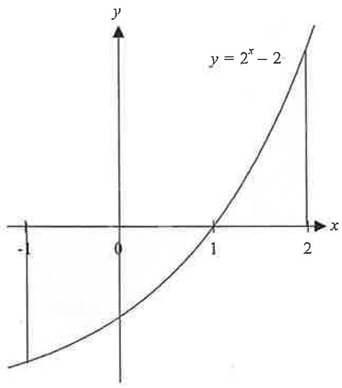
$$= \frac{1}{2}\left[x + \frac{\sin(4x)}{2}\right]_0^{\frac{\pi}{2}} = \frac{\pi}{4} \quad ✓$$

12 Find the volume of the solid of revolution formed when the graph of $y = \sqrt{\frac{1+2x}{1+x^2}}$ is rotated around the x-axis over the interval [0, 1]. 2

$$V = \pi\int_0^1 \left(\sqrt{\frac{1+2x}{1+x^2}}\right)^2 dx = \pi\int_0^1 \left(\frac{1}{1+x^2} + \frac{2x}{1+x^2}\right) dx$$

$$V = \pi\left[\arctan x + \ln\left(1+x^2\right)\right]_0^1 = \pi\left(\frac{\pi}{4} + \ln 2\right) u^3$$

13 In the diagram, the region bounded by the cyrve $y = 2^x - 2$ and the $x -$ axis between $x = -1$ and $x = 2$ is rotated through one revolution about the $x -$ axis. Find in simplest exact form the volume of the solid formed. 2



$$\int_{-1}^{2} \pi\left(2^x - 2\right)^2 dx = \pi\int_{-1}^{2}\left(2^{2x} - 2\times 2^x + 4\right) dx$$

$$= \pi\int_{-1}^{2}\left(e^{2x\ln 2} - 2e^{x\ln 2} + 4\right) dx$$

$$= \pi\left[\tfrac{1}{2\ln 2} e^{2x\ln 2} - \tfrac{2}{\ln 2} e^{x\ln 2} + 4x\right]_{-1}^{2}$$

$$= \pi\left[\tfrac{1}{2\ln 2} 2^{2x} - \tfrac{2}{\ln 2} 2^x + 4x\right]_{-1}^{2}$$

$$= \pi\left\{\tfrac{1}{2\ln 2}\left(16 - \tfrac{1}{4}\right) - \tfrac{2}{\ln 2}\left(4 - \tfrac{1}{2}\right) + 4\left(2+1\right)\right\}$$

$$= \pi\left(\tfrac{7}{8\ln 2} + 12\right)$$